

CSE 405: Computer Security

Class Test 2 – May 27 2025

Duration: 25 minutes

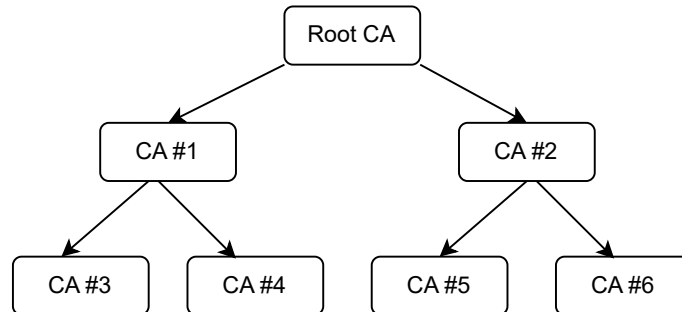
Total Marks: 20

Name: Mallory

ID: 31337

Multiple Choice Questions

$5 \times 2 = 10$ marks



1. Suppose CA#4 issues a certificate for Alice. Which of the following are included in Alice's certificate? You may check multiple boxes if applicable.
 - ☐ Alice's private key
 - ☒ Alice's public key
 - ☐ A signature on Alice's private key, signed by CA#4's private key
 - ☒ A signature on Alice's public key, signed by CA#4's private key
 - ☐ A signature on Alice's public key, signed by Alice's private key
2. Assume that you only trust the Root CA's public key. Which of following certificates do you need to verify the authenticity of Alice's certificate? You may check multiple boxes if applicable.
 - ☒ Alice's certificate
 - ☒ CA #1's certificate
 - ☐ CA #2's certificate
 - ☐ CA #3's certificate
 - ☒ CA #4's certificate
3. Assume that there is a central Trusted Directory for distributing certificates. Alice and Bob have exchanged and verified their public keys using this directory. Which of the following statement is **incorrect**? (Either B or C can be correct. Once Alice and Bob have verified their public keys, they can communicate securely without the directory.)
 - A. The trusted directory has scalability issues.
 - ✓B. If the directory stops working, Alice and Bob cannot communicate securely.
 - ✓C. If Mallory hacks the directory, she can impersonate anyone registered in the directory.
 - B. The trusted directory is a single point of failure.
4. Suppose a database stores: $H(H(\text{password} \parallel \text{salt}_1) \parallel \text{salt}_2)$, salt_1 and salt_2 . Given N users and M possible password permutations, what is the time complexity of cracking all passwords?
 - ✓A. $O(N \cdot M)$
 - B. $O(N + M)$
 - C. $O(N \cdot M^2)$
 - D. $O(N + M^2)$
5. Alice wants to send a message M to Bob. She sends the ciphertext C encrypted with RSA and the signature S of the message (also signed with RSA) to Bob. Which of the following statement is true?
 - A. This is secure as long as Alice uses a strong RSA key, preferably 2048 bits or more.
 - B. Mallory can forge the signature S for any message and impersonate Alice.
 - ✓C. Mallory can decrypt the ciphertext C and read the message M without having access to Alice's private key.
 - C. Mallory can multiply C and S to recover the original message M .

Recall the definition of ElGamal encryption:

- b is the private key, and $B = g^b \bmod p$ is the public key.
- $\text{Enc}(B, M) = (C_1, C_2)$, where $C_1 = g^r \bmod p$ and $C_2 = M \times B^r \bmod p$
- $\text{Dec}(b, C_1, C_2) = C_1^{-b} \times C_2 \bmod p$

Is it possible for Mallory to modify the ciphertext (C_1, C_2) to produce a different plaintext message M' (not necessarily of her choosing) without knowing the private key b ? Justify with necessary equations. **3 marks**

There are multiple ways to achieve this, one possible way is to set $C'_2 = C_2 \cdot k$, where k can be any integer. Then, the modified ciphertext becomes (C_1, C'_2) . The decryption process will yield a different plaintext message M' as follows:

$$M' = C_1^{-b} \cdot C'_2 \bmod p = C_1^{-b} \cdot (C_2 \cdot k) \bmod p = g^{-br} \cdot (M \cdot g^{br} \cdot k) \bmod p = M \cdot k \bmod p$$

Let $p = 17$ and $q = 11$ be two prime numbers for RSA. Compute the following:

1. n and $\phi(n)$ **2 marks**
2. Choose $e = 7$ as the public key. Compute the corresponding private key d . **2 marks**
3. Encrypt the message $m = 9$ using the private key d . **3 marks**

Answers:

1. $n = p \cdot q = 17 \cdot 11 = 187$ and $\phi(n) = (p - 1)(q - 1) = (17 - 1)(11 - 1) = 16 \cdot 10 = 160$
2. To find d , we need to solve $d \cdot e \equiv 1 \bmod \phi(n)$. Thus, we need to find d such that $d \cdot 7 \equiv 1 \bmod 160$. Using the Extended Euclidean Algorithm, we find $d = 23$.
3. To encrypt the message $m = 9$ using the private key $d = 23$, we compute $c = m^d \bmod n = 9^{23} \bmod 187$. The result is $c = 9^{23} \bmod 187 = 36$.

Note: Some of you may have encrypted the message using the public key e instead of the private key d . This is also acceptable, in that case the answer for part 3 would be $c = m^e \bmod n = 9^7 \bmod 187 = 70$.